

Module 3: The Central Concepts of Automata

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Alphabets, Strings and Languages

ALPHABETS & STRINGS

Formal Definitions & Operations

Alphabet

$\Sigma = \{a, b, c\}$

String

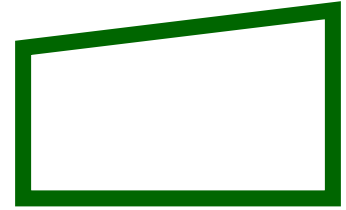
$w = \text{"abc"}$

Operations

$w_1 \circ w_2$

SYSTEM DESIGN SERIES

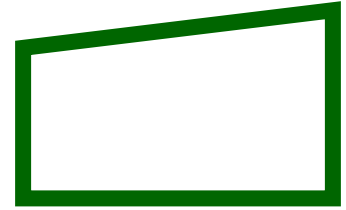
Alphabets



An alphabet is a finite, non-empty set of symbols:

- We use the symbol Σ (sigma) to denote an alphabet
- Examples:
 - Binary: $\Sigma = \{0,1\}$
 - All lower case letters: $\Sigma = \{a,b,c,..z\}$
 - Alphanumeric: $\Sigma = \{a-z, A-Z, 0-9\}$
 - Numeric: $\Sigma = \{0,1, 2, 3, .. 9\}$
 - DNA molecule letters: $\Sigma = \{A,C,G,T\}$
 - Initials of my name: $\Sigma = \{A, B, G\}$
 - Octal numbers: $\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7\}$
 - Arbitrary: $\Sigma = \{a, b, c\}$

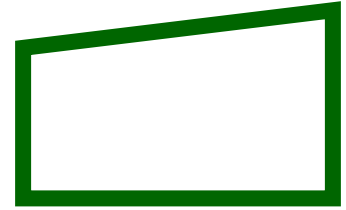
Strings



A string or word is a finite sequence of symbols chosen from Σ

- **Empty string is ε (or “epsilon”)**
 - Length of a string w , denoted by “ $|w|$ ”, is equal to the *number of (non- ε) characters in the string*
 - Let $\Sigma = \{0, 1\}$
 - *e.g., $x = 010100$ $|x| = 6$*
 - *$x = 01 \ \varepsilon \ 0 \ \varepsilon \ 1 \ \varepsilon \ 00 \ \varepsilon$ $|x| = 6$*
 - *$y = 111$ $|y| = 3$*
- xy** = concatenation of two strings x and y
- $x = 010100; y = 111$*
- $xy = 010100111$ $|xy| = 9$*

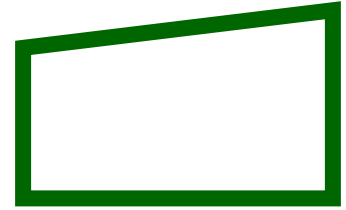
Powers of an alphabet



Let Σ be an alphabet.

- Σ^k = the set of all strings of length k
- If $\Sigma = \{0, 1\}$
- $\Sigma^1 = \{0, 1\}$
- $\Sigma^2 = \{01, 00, 10, 11\}$
- $\Sigma^3 = \{000, 111, 010, 101, 001, 100, 011, 110\}$
- $\Sigma^4 = \{0000, 1111, 0101, 1010, 0011, 1100, \dots\}$
- $\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$
- $\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots$

Powers of an alphabet



Let Σ be an alphabet.

- If $\Sigma = \{0, 1\}$
- $\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$
- $\Sigma^* = \{\epsilon, 0, 1, 00, 01, 10, 11, 000, 001, 010, \dots\}$
- $\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots$
- $\Sigma^+ = \{0, 1, 00, 01, 10, 11, 000, 001, 010, \dots\}$

Languages & Grammars

An **alphabet** is a set of symbols:

Or “**words**” $\{0,1\}$

Sentences are strings of symbols:

0,1,00,01,10,1,...

A **language** is a set of sentences:

$L = \{000,0100,0010,\dots\}$

A **grammar** is a finite list of rules defining a language.

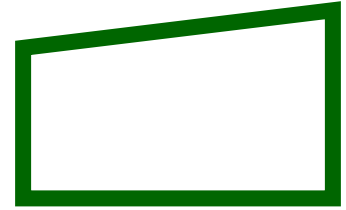
$S \longrightarrow 0A$ $B \longrightarrow 1B$

$A \longrightarrow 1A$ $B \longrightarrow 0F$

$A \longrightarrow 0B$ $F \longrightarrow \epsilon$

- Languages: “A language is a collection of sentences of finite length all constructed from a finite alphabet of symbols”
- Grammars: “A grammar can be regarded as a device that enumerates the sentences of a language” - nothing more, nothing less
- N. Chomsky, *Information and Control*, Vol 2, 1959

Languages



L is said to be a language over alphabet Σ , only if $L \subseteq \Sigma^$*

→ this is because Σ^* is the set of all strings (of all possible length including 0) over the given alphabet Σ

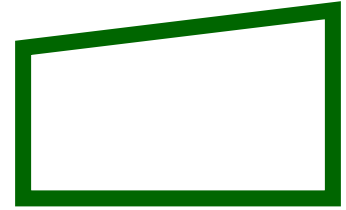
Examples:

1. Let L be *the* language of all strings consisting of n 0's followed by n 1's:
$$L = \{\epsilon, 01, 0011, 000111, \dots\}$$
2. Let L be *the* language of all strings with equal number of 0's and 1's:
$$L = \{\epsilon, 01, 10, 0011, 1100, 0101, 1010, 1001, \dots\}$$

Definition: $\emptyset$ denotes the Empty language

- Let $L = \{\epsilon\}$; Is $L = \emptyset$?

The Membership Problem



Given a string $w \in \Sigma^$ and a language L over Σ , decide whether or not $w \in L$.*

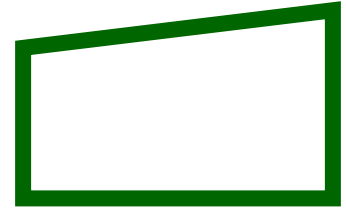
Example:

Let $w = 100011$

$|w| = 6$

Q: Is $w \in$ the language of strings with equal number of 0s and 1s? **YES!**

Some Applications



Software for designing and checking the behavior of digital circuits

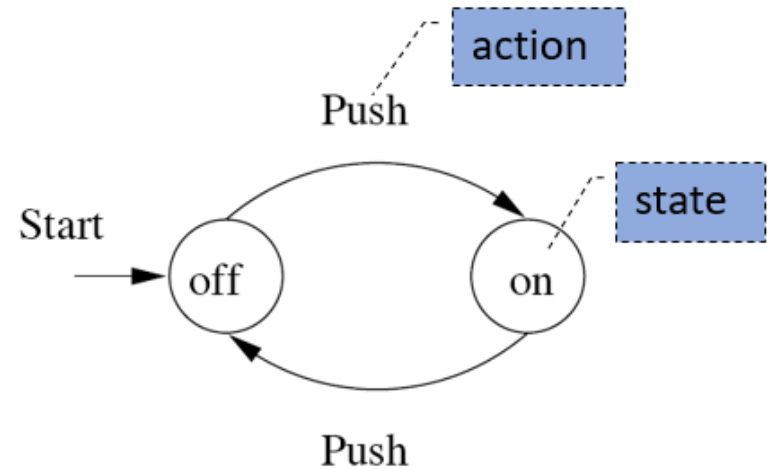
Lexical analyzer of a typical compiler.

Software for scanning large bodies of text (e.g., web pages) for pattern finding

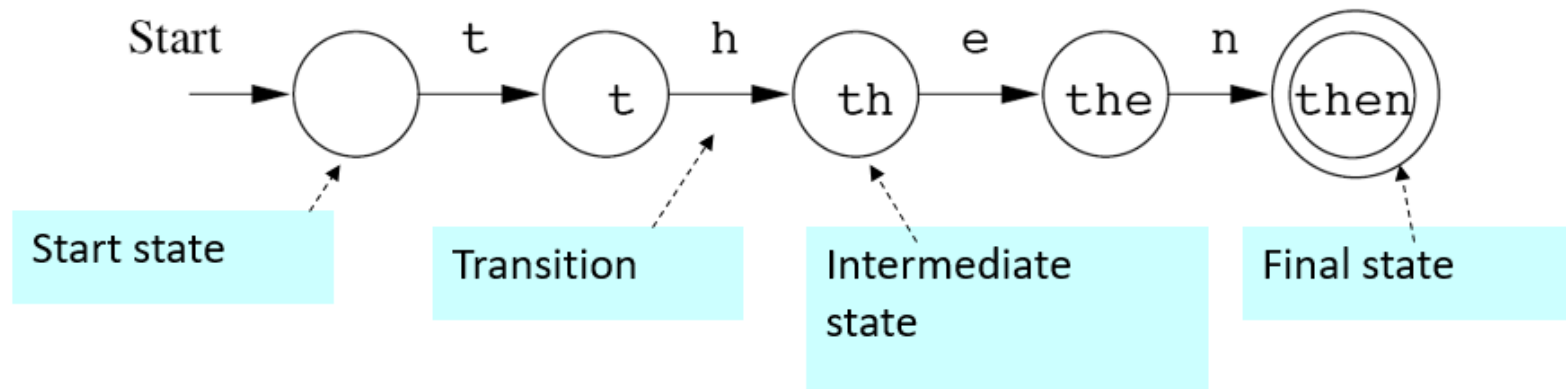
Software for verifying systems of all types that have a finite number of states (e.g., stock market transaction, communication/network protocol)

Finite Automata Examples

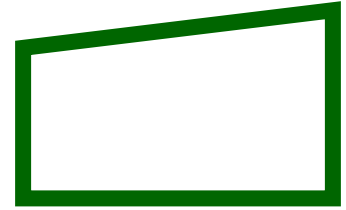
- On/Off switch



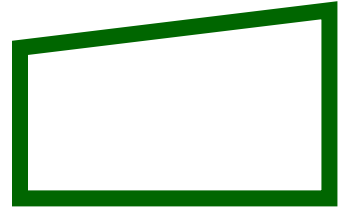
- Modeling recognition of the word "then"



Takeaways



- Defined alphabet $\rightarrow \Sigma$
- Defined string $\rightarrow$ sequence of symbols in the alphabet.
- Defined languages $\rightarrow L$.
- Computed the length of a string.
- Concatenation
- Powers of an alphabet
- The empty string
- Applications of automata
- Some examples of finite automata.



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